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Stock Market Prediction

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Chapter 1: Introduction

1.1 Background and Overview

Stock market prediction has long been a complex yet attractive area for financial institutions and individual investors. It involves anticipating future movements in stock prices to make informed trading decisions. In this research, we explore two separate AI-based models applied to the Egyptian stock market, specifically focusing on the **EGX-30 index**.

The first model utilizes deep learning (**LSTM**) with a technical analysis foundation, relying on historical stock data and key indicators such as **RSI**, **MACD**, Stochastic, **ATR**, **ADX**, and **ROC**. These indicators are selected based on their theoretical relevance and proven efficacy in identifying trading opportunities. This model aims to classify market conditions into actionable signals: **Buy**, **Hold**, or **Sell**.

The second model leverages **Natural Language Processing (NLP)** to perform sentiment analysis on Arabic financial news using a fine-tuned **CAMeL BERT** model. The sentiment output (**Positive**, **Neutral**, or **Negative**) provides an independent viewpoint on investor mood and economic perception.

And in this research, we explore multiple AI and statistical models applied to the Egyptian stock market, specifically focusing on the EGX-30 index.

The first branch includes **traditional time-series forecasting models** such as ARIMA, SARIMA, and SARIMAX. These models provide interpretable predictions and are effective for modeling linear trends, especially in stable conditions.

The second branch includes **deep learning models** like LSTM and Bidirectional LSTM, which are designed to capture complex, non-linear temporal dependencies in stock prices. Additionally, a **hybrid ARIMA-LSTM approach** was implemented to combine linear and nonlinear modeling strengths.

These models work independently, but together they offer a multi-dimensional view of the Egyptian market—covering historical trends, nonlinear dynamics, and sentiment analysis.

1.2 Problem Statement

Predicting financial markets is inherently uncertain due to their volatile and multi-factorial nature. Traditional statistical models fail to account for nonlinear patterns, and most deep learning approaches focus on raw price prediction, which can be misleading. Additionally, the influence of market sentiment is often ignored despite its measurable impact on stock prices.

This research addresses two core problems:

- How to develop a reliable signal-based prediction model using historical EGX-30 data.
- How to automatically assess the sentiment of Arabic financial news to support market analysis.

1.3 Objectives

Build an LSTM model that generates Buy/Sell/Hold signals based on historical data and technical indicators.

Fine-tune a CAMEL BERT model to classify Arabic financial news into sentiment categories.

Implement **ARIMA**, **SARIMA**, and **SARIMAX** models to forecast price trends using linear and seasonal components.

Build an **LSTM regression model** to predict future stock prices based on historical indicators.

Construct a **Bidirectional LSTM (BiLSTM)** model to capture past and future dependencies in stock time series data.

Develop a **hybrid ARIMA–LSTM model** that uses ARIMA to capture linear trends and LSTM to model nonlinear residuals

Evaluate each model using appropriate performance metrics such as RMSE, MAE, MAPE (for regression) and accuracy, precision, recall, and F1-score (for classification)

1.4 Scope

This project targets the Egyptian stock market, particularly the **EGX-30 index**, using historical data from **2008 to 2025**. The **forecasting models**—including ARIMA, SARIMA, SARIMAX, LSTM, Bidirectional LSTM, and Hybrid ARIMA–LSTM—were trained and evaluated on stock data from **2020 to 2025**.

The **news sentiment analysis** portion of the project focuses exclusively on **Arabic-language financial news**, collected from verified sources. News content is not translated or cross-referenced with English material.

Each model (forecasting and sentiment analysis) operates **independently** and was evaluated separately using relevant metrics. There is **no integration** between the LSTM forecasting models and the CAMEL BERT sentiment analysis in the current phase of the project.

Chapter 2: Literature Review

2.1 Overview

This chapter presents the theoretical background, surveys related work, and critically analyzes previous research on **three distinct domains**:

1. **Statistical time-series forecasting models** (ARIMA, SARIMA, SARIMAX), which form the foundation of classical financial modeling.
2. **Deep learning-based technical analysis**, particularly using models like LSTM and Bidirectional LSTM for capturing complex temporal dependencies in stock data.
3. **Arabic financial sentiment analysis using NLP**, focusing on transformer-based models such as CAMEL BERT.

Each modeling approach is explored independently in terms of its foundations, real-world applications, strengths, and limitations, with particular emphasis on relevance to the Egyptian market context.

2.2 LSTM-Based Forecasting

2.2.1 Background

Financial forecasting is a cornerstone of decision-making in capital markets. Investors, traders, and analysts continuously seek methods to predict market trends, identify profitable opportunities, and minimize risks. Traditional techniques such as technical and fundamental analysis have long been used for this purpose. However, with the surge in data availability and computational power, machine learning and deep learning models have emerged as powerful tools in financial prediction.

In particular models like Long Short-Term Memory (LSTM) networks and BERT-based transformers have gained traction due to their ability to model sequential patterns and extract sentiment from unstructured data, respectively. Combining both quantitative price patterns and qualitative sentiment analysis has shown promise in improving prediction performance, especially in dynamic and volatile markets.

2.2.2 Literature Survey

LSTM

Dhokane and Agarwal (2024) used LSTM with RSI and EMA indicators to forecast Indian stock prices, achieving improved accuracy.

Kamalov (2020) applied LSTM with RSI to detect major price shifts, outperforming classical models like Random Forest. Other studies integrated RSI with LSTM alongside indicators like MACD and Bollinger Bands, enhancing prediction performance in emerging markets.

Patel et al. (2015) utilized SVM and decision tree models with technical indicators to forecast Indian stock movements.

Fischer and Krauss (2018) implemented LSTM networks for S&P 500 prediction and achieved significant gains over traditional models.

BERT

XGBoost + TF-IDF – classical ML with Arabic financial headlines

AraBERT + Lexicon – Arabic lexicon-based sentiment classification

Xing et al. (2018) explored CNN-based sentiment models trained on news headlines for market forecasting.

Araci (2020) introduced FinBERT, fine-tuned for English-language financial sentiment tasks. X

CAMeL BERT, trained on Arabic corpora, has shown strong results in general NLP benchmarks but hasn't been applied to financial sentiment tasks.

Challenges in LSTM-Based Forecasting

Most studies focused on price regression rather than signal classification. Additionally, they were developed for mature markets (e.g., U.S., China, India) and ignored emerging economies like Egypt. They also lacked long historical.

Unlike many prior studies that focus on raw price forecasting or binary up/down direction classification, our LSTM model directly outputs actionable trading signals (Buy, Hold, Sell). This approach reflects real-world decision-making needs and allows for more intuitive evaluation using classification metrics such as accuracy, precision, and F1-score.

Despite their promise, LSTM models face common challenges in financial prediction: Overfitting on noisy market data due to the non-stationary nature of time series.

Limited ability to incorporate exogenous information, such as news or economic indicators.

Sensitivity to window size and hyperparameters. Lack of semantic understanding — they treat inputs as numeric sequences and miss nuanced insights in natural language data.



2.3 Comparative Analysis of Related Work

Table for comparative Analysis of sentiment analysis models for financial news

Project/Study	Language	Data	Accuracy	Unique Points
Camel BERT for EGX Financial News	Arabic	Real Egyptian financial news	82%	Fine-tuning on Egyptian data; negative, neutral, positive labels
Camel BERT (General - Not Financial Specific)	Arabic	Generic/ non-news data	60–70% ~ 0.65	Weaker performance without financial fine-tuning
AraBERT + Lexicon	Arabic	Lexicon words	~ 55-65% Low	Failure to understand context; weak performance
FinBERT (English)	English	US market data	84–88% 0.82–0.85	Great performance, but not supports Arabic language
XGBoost + TF-IDF	Arabic	Financial headlines	~ 60-68% Low	The approach struggle with complexity of Arabic

LSTM Model Comparison with Evaluation Type

Study/language	Indicators / Inputs	Market / Dataset	Accuracy	Evaluation Type
Our LSTM Model	RSI, MACD, Stochastic, ATR, ADX, ROC	EGX-30 (Egypt)	64.71%	Buy/Hold/Sell Classification
Fischer et al. (Chinese)	Raw market data	China stock data	51.4% - 53.8%	Directional Forecasting
LSTM + Attention	Raw + Attention	Two Sigma (US)	~60%	Directional Forecasting
Reddit LSTM	3 Reddit features	Reddit dataset	~50%	Binary Classification
Baseline LSTM	Raw OHLC only	S&P 500 (simulated)	~50%	Price Regression
Dhokane & English	RSI, EMA	NSE (India)	Not reported	Price Forecasting
Kamal	RSI	Unspecified	Not reported	Major Price Shifts

Our work differs by:

- Targeting EGX-30, an emerging market index.
- Focusing on multi-indicator signal classification (Buy/Hold/Sell).
- Using historical data spanning 2008–2025.
- Existing Arabic NLP research rarely addresses financial sentiment. Most models either focus on English-language financial corpora or use lexicon-based approaches. To the best of our knowledge, no public dataset exists for Arabic financial sentiment.
- We created a labeled dataset from real Egyptian financial news.
- Although **CAMEL BERT** was not originally designed to understand financial or stock market content, we fine-tuned it to classify sentiment into Positive, Neutral, or Negative.

- The sentiment model is used as an independent decision-support tool.

Our Contribution and Advantages To address the above limitations, our proposed approach integrates the strengths of LSTM with advanced NLP models:

Hybrid Indicator Integration: Technical indicators like RSI and MACD are used as explicit features alongside LSTM inputs, supporting momentum-based pattern detection.

Robust Data Cleaning and Feature Engineering: Addressing overfitting by applying normalization, dropout, and selective feature extraction. This hybrid design not only leverages the sequential learning capacity of LSTM but also enhances its prediction quality through context-aware sentiment signals, making it more resilient to market volatility and news-driven fluctuations.

To the best of our knowledge, we are the first to perform **sentiment analysis** on a dataset specifically composed of **Egyptian financial news**, collected from **Mubasher Misr**, a leading source for stock market and economic news in Egypt.

2.4 Summary of Gaps and Contributions

Many forecasting models ignore sentiment, and most sentiment models ignore Arabic content.

Few applications target EGX or similar emerging markets.

This project applies LSTM for signal-based forecasting and CAMEL BERT for Arabic sentiment, each evaluated independently.

The novelty lies in the contextual adaptation to the Egyptian market and the dual-track analysis framework.

2.3 Time-Series Forecasting Using Classical Statistical Models

2.3.1 Background

Statistical methods such as **ARIMA** (AutoRegressive Integrated Moving Average), **SARIMA** (Seasonal ARIMA), and **SARIMAX** (ARIMA with exogenous variables) have been widely used for forecasting financial time series. These models are valued for their mathematical transparency and ability to capture linear patterns in stationary or seasonally-adjusted data.

While powerful for trend-based markets, classical models often struggle with high volatility or abrupt nonlinear shifts common in emerging markets. Recent studies have proposed hybrid approaches where statistical models capture general structure, and machine learning models account for residual noise.

2.3.2 Literature Survey

- **Box & Jenkins (1976)** introduced ARIMA modeling as a statistical foundation for time-series forecasting.
- **Contreras et al. (2003)** applied SARIMA to electricity prices and highlighted its strength in periodic trends.
- **Wang et al. (2021)** combined SARIMA and external variables (SARIMAX) to model exchange rates and commodity prices.
- Studies like **Ahmed et al. (2022)** explored ARIMA-SVR and ARIMA-LSTM hybrids for more adaptive modeling of volatile stock indices.

2.3.3 Challenges

- Classical models assume stationarity and linearity, which limit performance in highly dynamic financial markets.
 - They do not adapt well to external shocks unless exogenous variables are explicitly introduced.
 - Seasonal components are manually defined, and model tuning (e.g., p , d , q selection) can be computationally expensive without automation.
-

2.4 Deep Learning for Technical Forecasting

2.4.1 Background

Deep learning models, especially **Long Short-Term Memory (LSTM)** networks, are well-suited for sequential data like stock prices due to their ability to retain long-term temporal dependencies. **Bidirectional LSTM (BiLSTM)** extends this by processing sequences both forward and backward.

2.4.2 Literature Survey

- **Dhokane & Agarwal (2024)** used LSTM with RSI and EMA to predict Indian stock prices.
 - **Kamalov (2020)** applied LSTM + RSI to identify directional shifts in emerging markets.
 - **Fischer & Krauss (2018)** found that LSTM outperformed traditional models for S&P 500 forecasting.
 - **Recent studies** have introduced **hybrid LSTM–ARIMA** models, where ARIMA models general patterns and LSTM learns residuals.
-

2.5 Comparative Insights and Contributions

2.5.1 Summary of Gaps

- Most statistical models lack non-linear learning capacity.
- Deep learning models may overfit on noise and require large datasets and tuning.
- Arabic sentiment analysis remains underexplored, especially for financial contexts.

2.5.2 Our Work Differs By:

- Implementing and evaluating a **diverse set of models**: ARIMA, SARIMA, SARIMAX, LSTM, BiLSTM, and ARIMA–LSTM hybrids.
- Applying these models specifically to the **EGX-30**, an underrepresented emerging market.
- Using **technical indicators** (RSI, MACD, ATR, etc.) as features rather than raw prices.
- Evaluating models with regression metrics (RMSE, MAE, MAPE) and discussing practical forecasting implications.
- Including a **CAMeL BERT-based sentiment analyzer** (covered separately) for qualitative insight.

2.5.3 Summary

This chapter has reviewed a wide spectrum of financial forecasting approaches. Our contribution lies in:

- Contextual adaptation to **Egyptian market data**.
- Applying and comparing **both traditional and modern AI models**.
- Highlighting the strengths and weaknesses of each model family.

Chapter 3: Methodology

3.1 Overview

This chapter outlines the methodologies adopted to build and evaluate a comprehensive set of models for analyzing the Egyptian stock market (EGX-30 index). The methods fall into **three main categories**, each addressing a specific analytical dimension:

1. Time-Series Forecasting Models

- **Statistical models:** ARIMA, SARIMA, and SARIMAX are used to capture linear trends and seasonal patterns in historical stock price data.
- **Deep learning regression models:** These include a unidirectional LSTM, Bidirectional LSTM, and a Hybrid ARIMA–LSTM model, all trained to forecast future stock prices using technical indicators as input.

2. Technical Signal Classification Model

- A deep learning model based on **LSTM** is developed to generate actionable trading signals (Buy, Hold, Sell) based on engineered technical indicators derived from historical EGX-30 data. This classification approach aligns closely with decision-making needs in financial trading.

3. Sentiment Analysis Model

- A fine-tuned **CAMEL BERT model** is used for sentiment classification of Arabic financial news. It categorizes news content into Positive, Neutral, or Negative sentiment, providing a qualitative assessment of market psychology.

All models are developed, trained, and evaluated **independently** using appropriate performance metrics for their respective tasks (e.g., RMSE, MAE, MAPE for forecasting; accuracy, F1-score for classification).

3.2 LSTM-Based Technical Signal Classification

Data Collection and Preprocessing

The dataset consists of EGX-30 daily market data from 2008 to 2025. Each record includes Open, High, Low, Close, and Volume prices. From this raw data,

CSV loaded successfully:

	Date	Price	Open	High	Low	Vol.	Change %
0	2/6/2025	30,011.84	29,735.59	30,011.84	29,735.59	83.50M	0.93%
1	2/5/2025	29,735.59	29,668.47	29,772.63	29,596.50	103.71M	0.23%
2	2/4/2025	29,668.47	29,754.77	29,937.39	29,668.47	135.05M	-0.29%
3	2/3/2025	29,754.77	29,917.96	30,001.23	29,698.79	152.06M	-0.55%
4	2/2/2025	29,917.96	30,010.62	30,198.22	29,917.96	103.16M	-0.31%

We derived several technical indicators using sliding windows and rolling averages, including:

- Relative Strength Index (RSI)
- Moving Average Convergence Divergence (MACD)
- Stochastic Oscillator (%K, %D)
- Average True Range (ATR)

- Average Directional Index (ADX)
- Rate of Change (ROC)

Signal Labelling

A rule-based system was implemented to assign one of three trading signals (Buy = 1, Hold = 0, Sell = -1) based on indicator thresholds. For example:

- $RSI < 30 \rightarrow \text{Buy}$
- $RSI > 70 \rightarrow \text{Sell}$
- $ADX < 20 \rightarrow \text{Hold (weak trend)}$

Model Architecture

the LSTM model architecture includes:

Input: sequences of technical indicators over 10-day windows.

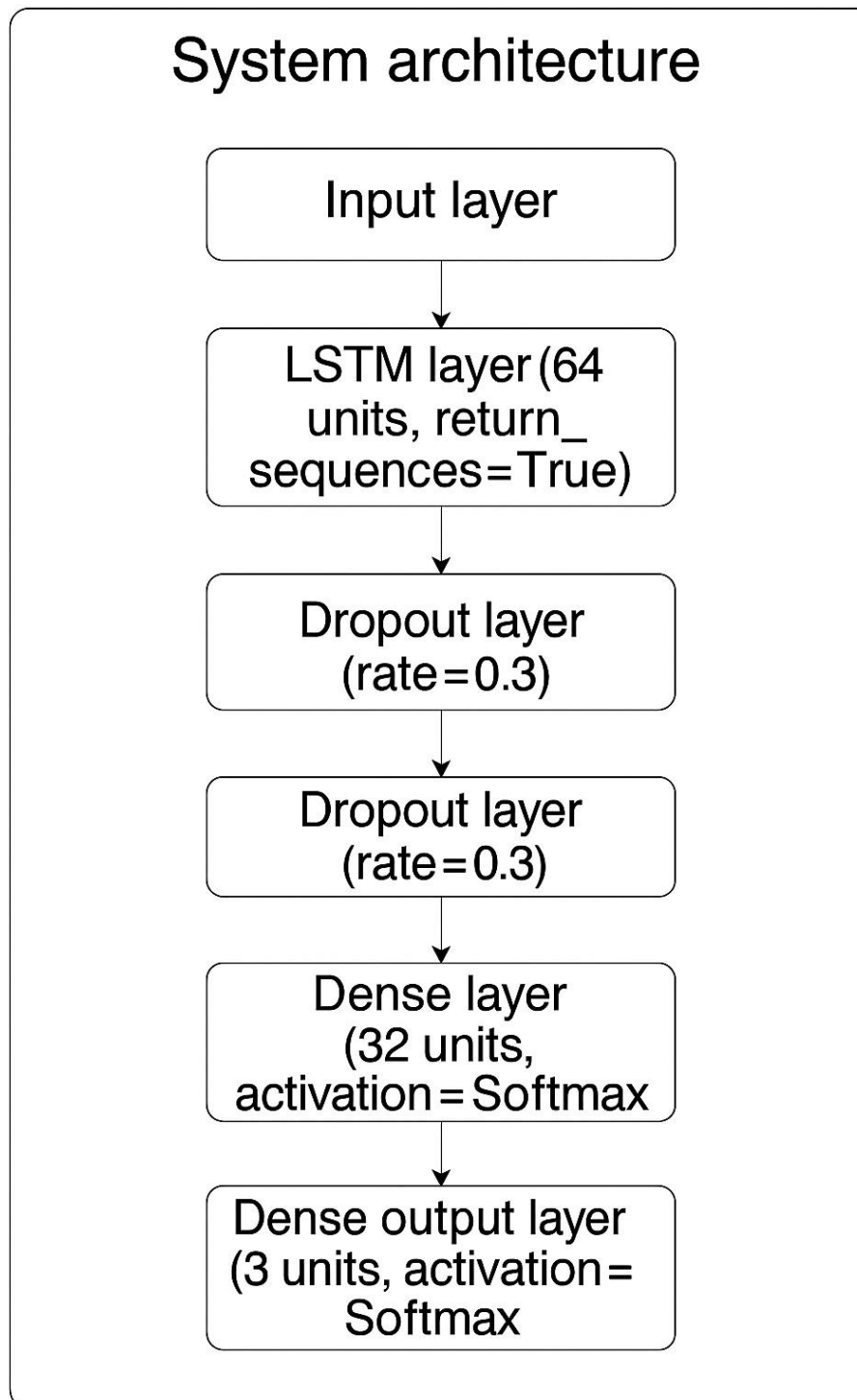
Layers: LSTM layer with 50 units $\rightarrow$ Dropout layer $\rightarrow$ Dense Softmax layer.

Output:

- 3-class classification (Buy, Hold, Sell)
- 3.2.4 Training and Evaluation
- Loss Function: Categorical Cross-Entropy
- Optimizer: Adam
- Epochs: 25
- Metrics: Accuracy, Precision, Recall, F1-score, and Confusion Matrix
- Train/Test Split: 80/20 with stratified sampling

Neural Network Architecture

The architecture of a neural network model, consisting of LSTM, Dropout, and Dense layers. Each layer's configuration, including type, trainable status, parameters, and weights, is provided below in a structured format.



Notes

- The total number of parameters in the model is the sum of parameters from all layers: 18,688 (LSTM) + 0 (Dropout) + 12,416 (LSTM) + 0 (Dropout) + 1,056 (Dense) + 99 (Dense) = 32,259.
- Shape information could not be retrieved for any layer, as indicated by the error: "‘LSTM’/‘Dropout’/‘Dense’ object has no attribute ‘input_shape’."

3.3 NLP-Based Sentiment Classification Using CAMEL BERT

Data Collection and Annotation Arabic financial news was collected from verified sources, cleaned, and labeled manually into three sentiment classes: Positive, Neutral, and Negative. Only formal financial language was included to ensure consistency.

5	المؤشر العام	positive
6	أعلنت أوراس	negative
7	أصدرت هيئة	neutral
8	نمو في إيرادات	positive
9	إعلان إفلاس	negative
10	اجتمعت إدارة	neutral
11	ارتفعت أرباح	positive
12	أعلنت مدينة	negative
13	عقدت وزارة	neutral
14	تم توزيع أرباح	positive
15	أعلن البنك	negative
16	تم توقيع مذكرة	neutral
17	سجل سهم	positive
18	إعلان إفلاس	negative
19	تم توقيع مذكرة	neutral
20	أعلنت جهينا	positive

To perform effective fine-tuning, we first trained the CAMEL BERT model on a **custom-labeled dataset** specifically designed to teach the model key financial and stock market terminology. This step was essential, as the base model lacks native understanding of domain-specific expressions used in Arabic financial contexts.

After fine-tuning, we applied the trained model to **real-world financial news** scraped from **Mubasher Misr**, a leading Egyptian stock market news platform. The scraped articles were cleaned, preprocessed, and then fed into the model to classify sentiment across a wide range of authentic market scenarios.

This dataset contains around 2,400 real news articles about the Egyptian stock market

				مباشرة
				قالت الهيئة العامة
				للرقابة المالية، إنه
				ليس لديها مانع بصفة
				مبدئية من قيام شركة
				بالتون القابضة
				بنشر دعوة لقدامى
				المساهمين للاكتتاب في
				الأسهم المتبقية البالغ
				عدد 424.22
				مليون سهم بقيمة
				اسمية 2 جنية للسهم
				وبذات شروط
76	الرقابة المالية	https://www.mubasher-misr.com	26 مارس 2025	أخبار الشركة

Preprocessing Steps

- Normalization of Arabic text
- Removal of URLs, punctuation, and repeated characters
- Tokenization using CAMEL BERT tokenizer

Model Setup and Fine-Tuning

- Base Model: CAMEL-Lab/bert-base-arabic-camelbert-mix
- Number of Classes: 3
- Framework: Hugging Face Transformers
- Training Strategy: Weighted loss for imbalance, early stopping, evaluation on validation set

Training Configuration

- Learning Rate: $2e-5$
- Batch Size: 16
- Epochs: 3
- Evaluation Metric: Macro F1-score, Label-wise Precision & Recall

Model Usage

The trained model classifies unseen news headlines into sentiment labels with confidence scores. It is deployed independently and is not integrated with the LSTM signal output.

3.4 Statistical and Deep Learning Regression Models

Data Collection and Preprocessing

The dataset includes **daily EGX-30 stock market data** from **2020 to 2025**. Each record contains the following fields:

- Open
- High
- Low
- Close
- Volume

From this raw data, several **technical indicators** were computed and used as features for the deep learning models:

- Relative Strength Index (RSI)
- Average True Range (ATR)
- Moving Average Convergence Divergence (MACD)
- Stochastic Oscillator (%K, %D)

For ARIMA-based models, no external features were required in the base case, but for SARIMAX, **technical indicators were included as exogenous variables**.

Model Architectures and Configurations

ARIMA / SARIMA / SARIMAX

- **ARIMA(3,2,3)** was selected using grid search with the lowest AIC value.
- **SARIMA(5,2,0)(0,0,0)[0]** modeled seasonal trends.
- **SARIMAX** was implemented using technical indicators as exogenous regressors.

All models were evaluated based on one-step-ahead forecasting using the test set.

LSTM Regression Model

- Input: Sequences of past prices and indicators over fixed time windows.
- Architecture:
 - LSTM (100 units, return_sequences=True)
 - Dropout (0.3)
 - LSTM (100 units)
 - Dropout (0.3)
 - Dense (1 unit for price prediction)
- Loss Function: Mean Squared Error
- Optimizer: Adam
- Training: 1000 epochs with early stopping and learning rate reduction.

Bidirectional LSTM (BiLSTM)

- Input: Sequences of historical data shaped for bidirectional temporal context.
 - Architecture:
 - BiLSTM (128 units, return_sequences=True) → LayerNormalization
 - BiLSTM (64 units, return_sequences=True) → LayerNormalization
 - BiLSTM (32 units) → LayerNormalization
 - Dense (1 unit)
 - Loss Function: Mean Squared Error
 - Optimizer: Adam
 - Training Strategy: Same as LSTM
-

Hybrid ARIMA–LSTM Model

- **ARIMA** was first trained to model and forecast stock price trends.
 - The **residuals** (difference between ARIMA predictions and actual prices) were calculated.
 - These residuals were then used as targets to train an LSTM model with technical indicators as input.
-

LSTM for residuals:

- Architecture: Same as standard LSTM regression
- Input: Sequences of RSI, ATR, MACD, Volume
- Target: Scaled ARIMA residuals

Training and Evaluation

All regression models were trained and tested independently. Evaluation metrics used:

- **Root Mean Squared Error (RMSE)**
- **Mean Absolute Error (MAE)**
- **Mean Absolute Percentage Error (MAPE)**

Train/test split: 80/20

Each model was evaluated based on its predictive accuracy on the test set, and comparisons were made across all models to identify strengths and limitations.

3.5 Summary

This chapter presented the methodologies used to develop and evaluate a range of models for stock market analysis targeting the EGX-30 index. The approaches include:

- **Statistical forecasting models** (ARIMA, SARIMA, SARIMAX) for capturing linear and seasonal trends in stock price data.
- **Deep learning regression models** (LSTM, Bidirectional LSTM, Hybrid ARIMA–LSTM) for modeling nonlinear relationships and improving prediction accuracy.
- **LSTM-based classification model** for generating actionable trading signals (Buy, Hold, Sell) from historical technical indicators.
- **CAMeL BERT sentiment analysis model** for classifying Arabic financial news headlines into sentiment categories to reflect public mood and market perception.

All models were trained, validated, and evaluated **independently** using appropriate performance metrics for their respective tasks (e.g., RMSE, MAE, MAPE for regression; accuracy and F1-score for classification).

The next chapter will describe the system architecture, tools, and implementation pipeline for each model in detail.

Chapter 4: System Design and Implementation

4.1 Overview

This chapter outlines the design and implementation of all models developed in this project for analyzing the Egyptian stock market (EGX-30). The models are grouped into three main systems based on their function and architecture:

- **Time-Series Forecasting Systems:** These include the implementation of classical statistical models such as ARIMA, SARIMA, and SARIMAX, as well as deep learning regression models like LSTM, Bidirectional LSTM, and a hybrid ARIMA–LSTM model. Each of these models is designed to predict future stock prices using historical data and technical indicators.
- **Technical Signal Classification System:** A deep learning-based LSTM classifier was implemented to generate trading signals (Buy, Hold, Sell) by learning temporal patterns from sequences of technical indicators.
- **Sentiment Analysis System:** A fine-tuned CAMEL BERT model was built to classify Arabic financial news headlines into Positive, Neutral, or Negative sentiment. This system operates independently and serves as a qualitative decision-support tool.

Each system is described independently, with details on architecture, implementation workflow, development environment, and toolkits used. The following sections provide in-depth descriptions and design considerations for each component.

4.2 LSTM Signal Prediction System

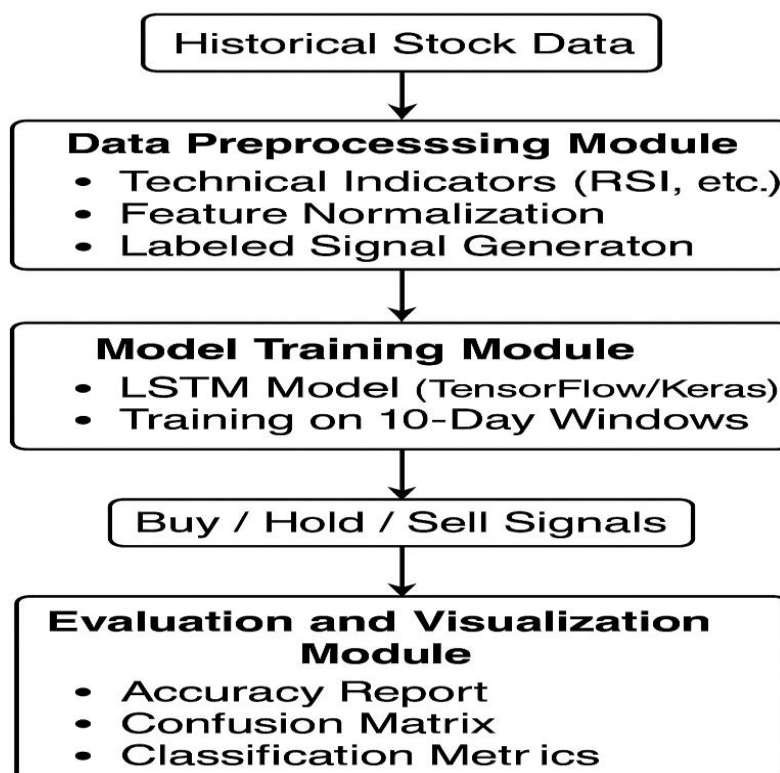
System Architecture

The LSTM-based system is composed of three major stages:

Data Preprocessing Module: Responsible for calculating technical indicators (RSI, MACD, etc.), normalizing features, and generating labeled signals.

Model Training Module: Implements the LSTM model using TensorFlow/Keras, trains on 10-day historical windows, and outputs Buy/Hold/Sell signals.

Evaluation and Visualization Module: Generates accuracy reports, confusion matrices, and classification metrics.



System Architecture

Software and Tools

Programming Language: Python 3.10

Libraries: NumPy, Pandas, Scikit-learn, TensorFlow, Matplotlib

Development Platform: Jupyter Notebook, Visual Studio Code

Deployment Environment

CPU/GPU: NVIDIA-enabled GPU (optional for faster training)

Operating System: Windows/Linux

Output: CSV file containing daily signals for each trading day

4.3 CAMEL BERT Sentiment Analysis System

System Architecture

The sentiment classification pipeline includes:

Domain Adaptation & Fine-Tuning Module:

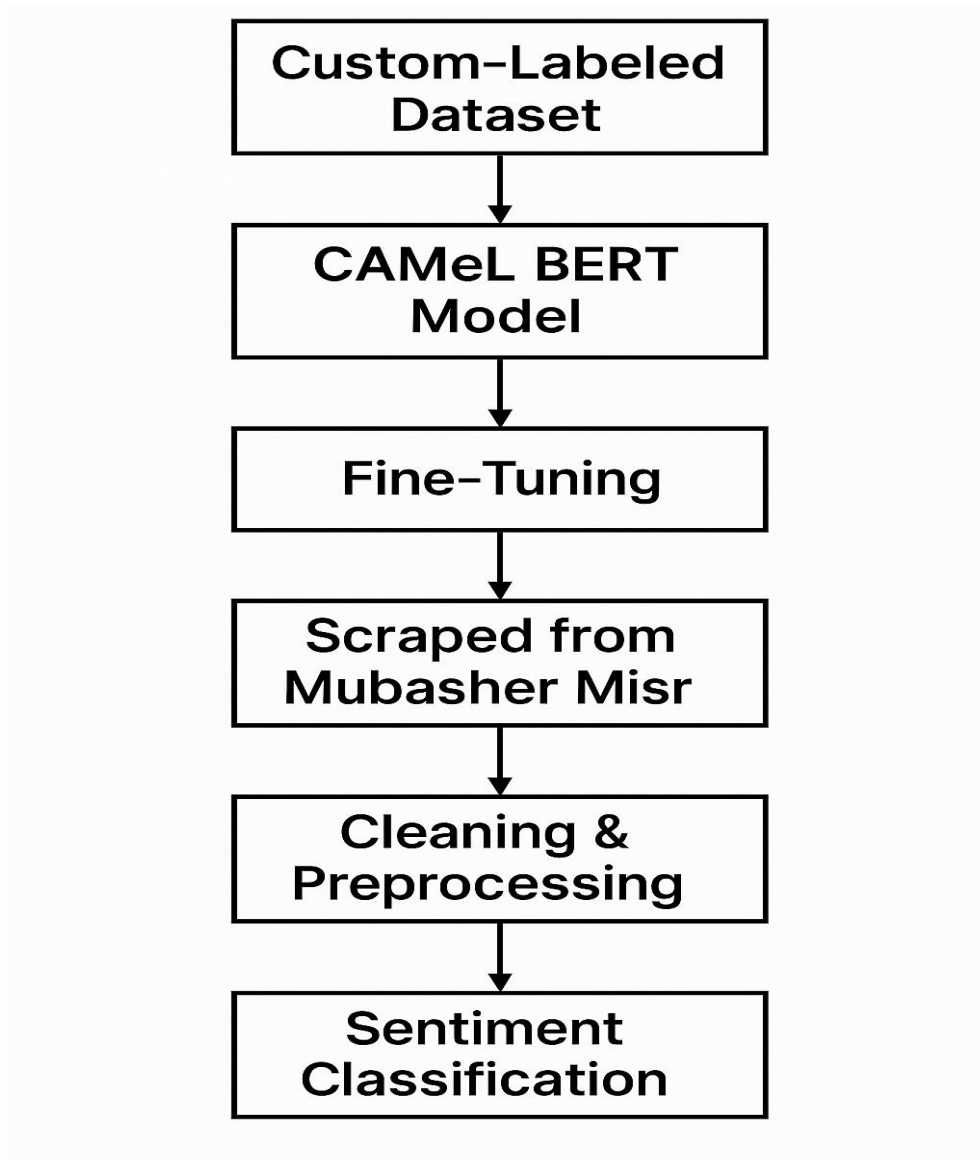
Begins with a custom-labeled dataset rich in Arabic financial and stock market terminology. This dataset is used to fine-tune the base CAMEL BERT model, enabling it to better understand domain-specific expressions that are not natively supported in the pre-trained version.

News Acquisition & Preprocessing Module:

Real-world Arabic financial news articles are scraped from *Mubasher Mizr*, a prominent Egyptian stock market news platform. The scraped articles undergo thorough cleaning and preprocessing—removing noise such as URLs, punctuation, and redundant characters—to ensure high-quality input for the model.

Model Inference Module:

The cleaned news articles are passed into the fine-tuned CAMEL BERT model to classify sentiment into **Positive**, **Neutral**, or **Negative** categories.



Result Aggregation Module:

Sentiment predictions, along with their confidence scores, are stored in structured formats such as CSV or Excel files for downstream analysis, visualization, or integration into decision-support tools

Software and Tools

Programming Language: Python 3.10

Libraries: Hugging Face Transformers, PyTorch, Pandas, Scikit-learn

Training Environment: Google Colab / Local GPU environment

Deployment Environment

Model Storage: trained_financial_sentiment_model directory

Output Format: CSV file with sentiment labels and confidence scores for each news item

Can be integrated into dashboards or monitoring tools if needed

4.4 Forecasting System: ARIMA, LSTM, BiLSTM, and Hybrid Models

System Architecture

The forecasting system consists of four independent modules, each developed to predict EGX-30 stock prices using different approaches: classical statistical models, deep learning models, and a hybrid technique combining both.

1. ARIMA-Based Statistical Forecasting

- **Data Handling Module:** Daily EGX-30 stock data from 2020–2025 is filtered, cleaned (handling missing values, formatting dates, and normalizing columns), and split into training and testing sets (80/20).

- **Model Training Module:**
 - ARIMA and SARIMA models are configured using both grid search and auto-arima for hyperparameter tuning (p , d , q).
 - SARIMAX integrates exogenous variables such as Volume, RSI, ATR, and MACD.
 - **Residual Analysis Module:**
 - Model residuals are analyzed using statistical plots: ACF, PACF, QQ plots, and KDEs to ensure white-noise behavior.
 - **Forecasting Module:**
 - Forecasts are generated and evaluated using RMSE, MAE, and MAPE.
 - Visualization tools plot actual vs. forecasted prices for interpretability.
-

2. Deep Learning Model: LSTM Regression

- **Data Preparation Module:**
 - Features such as RSI, ATR, MACD, and Volume are scaled using MinMaxScaler.
 - Time-windowed sequences (30-day lookbacks) are created for supervised learning.

- **Model Architecture:**
 - Two LSTM layers with 100 units each, regularized with L2 and dropout layers.
 - Output layer has one unit for continuous price prediction.
 - **Training Module:**
 - Optimizer: Adam
 - Loss: Mean Squared Error
 - Early stopping and learning rate reduction callbacks are used to stabilize training.
 - **Evaluation & Output:**
 - The model is validated on the test set using RMSE, MAE, and MAPE.
 - Predictions are inverse-transformed to actual price scale.
 - Results are visualized with plots comparing predicted vs. actual prices.
-

3. Hybrid ARIMA–LSTM Model

- **Integration Module:**
 - Residuals from the ARIMA model are computed and used as the target for a new LSTM model.
- **Feature Module:**
 - Technical indicators (Volume, ATR, RSI, MACD) serve as input features for residual modeling.

- Both inputs and residual targets are scaled and sequenced.
 - **LSTM Architecture:**
 - Two-layer LSTM with 100 units each, plus dropout and L2 regularization.
 - **Prediction Module:**
 - Forecasted ARIMA trend is combined with LSTM-predicted residuals to form the final hybrid forecast.
 - Performance is evaluated on the test set using standard regression metrics.
-

4. Bidirectional LSTM (BiLSTM)

- **Preprocessing Module:**
 - Uses the same time-sequenced features as LSTM but applies StandardScaler instead of MinMaxScaler.
- **Model Architecture:**
 - Three Bidirectional LSTM layers (128, 64, 32 units) with LayerNormalization after each.
 - Output layer has one unit for continuous regression.
 - Loss: Huber loss with Mean Absolute Percentage Error (MAPE) as a metric.

- **Training and Evaluation:**
 - Trained using early stopping and learning rate scheduling.
 - Final predictions are visualized and compared with actual prices.

Software and Tools

- **Programming Language:** Python 3.10
 - **Libraries:**
 - Statistical Models: statsmodels, pmdarima
 - Deep Learning: TensorFlow, Keras
 - Data Processing: NumPy, Pandas, Scikit-learn, ta
 - Visualization: Matplotlib, Seaborn
 - **Development Platform:** Jupyter Notebook, Visual Studio Code
-

Deployment Environment

- **Hardware:** CPU / optional GPU (for faster LSTM training)
- **OS:** Compatible with Windows and Linux
- **Output:** Forecasted price values, plots, and CSV exports with predictions and performance metrics

4.5 Data Flow and Workflow Integration Although the models are independent, they share a unified design workflow:

LSTM uses structured time-series data from EGX-30 .

CAMEL BERT uses unstructured Arabic text from financial news.

Both models write their output predictions to files for further decision-making or visualization.

4.6 Summary:

This chapter presented the complete system design for both models. While distinct in input format and architecture, they follow similar workflows in preprocessing, modeling, and output generation. The next chapter will present the results and performance analysis of each model.

In more detail:

- **LSTM Signal Prediction System** utilizes EGX-30 structured time-series data to learn technical trading signals using sequential dependencies. It includes modules for preprocessing (technical indicator generation), model training (deep LSTM layers with regularization), and evaluation (metrics and visual reports).
- **CAMEL BERT Sentiment Analysis System** processes unstructured Arabic financial news, using fine-tuned domain-specific BERT to classify sentiment. It features dedicated stages for web scraping, data cleaning, model inference, and result aggregation for integration into financial decision tools.

- **Hybrid ARIMA–LSTM Model** combines ARIMA’s strength in modeling linear components with LSTM’s ability to learn non-linear residual patterns. The hybrid setup significantly improves prediction accuracy by correcting ARIMA residuals through an LSTM layer trained on technical indicators.
- **Bidirectional LSTM** extends standard LSTM by incorporating both past and future temporal dependencies. This approach captures richer context around price behavior and enhances the model’s ability to anticipate market movements.

Although each model runs independently, they are unified by a common workflow architecture:

- Ingestion of structured (market data) or unstructured (news articles) sources.
- Specialized domain-aware preprocessing pipelines.
- Deep or statistical modeling stages.
- Prediction output written in standardized formats (e.g., CSV) for further decision-making or integration.

Together, this unified system design supports a comprehensive, multi-perspective market analysis framework combining quantitative and qualitative signals.

Chapter 5 Results

5.1 Overview

This chapter presents the experimental results and evaluation metrics for both the LSTM-based signal prediction model and the CAMEL BERT-based sentiment analysis model. Each model was assessed independently using appropriate classification metrics. Additionally, this chapter documents several alternative experimental approaches that were explored but discarded due to poor performance or misalignment with the problem domain.

5.2 LSTM Signal Prediction Results

Evaluation Metrics

The LSTM model was evaluated on a held-out test set (20% of the dataset) using the following metrics:

- Accuracy: 64.71%
- Precision (macro average): 0.70
- Recall (macro average): 0.80
- F1-score (macro average): 0.67

Confusion Matrix

The confusion matrix shows balanced performance across the three classes:

- Buy: High precision, occasional misclassification as Hold
- Hold: Slightly lower recall, often confused with Sell
- Sell: Good detection but slightly lower confidence

Performance Insights

- The model performed better on clear trend conditions (e.g., strong bullish or bearish signals).
- Ambiguous market phases led to confusion between Hold and other classes.
- The use of combined indicators improved class separability.

Preliminary Experiments and Lessons Learned

- Initially, a **CNN model** was tested for time-series forecasting. However, due to CNN's architecture being optimized for spatial feature detection (like in images), the model failed to understand temporal dependencies in financial data.
- A second attempt used **LSTM for regression**, predicting the exact next-day closing price. This approach proved impractical due to high volatility and unpredictable market noise.
- A **hybrid model combining LSTM and ARIMA** was also explored. The integration was unstable and produced inconsistent outputs, leading to the decision to rely solely on the LSTM classifier.

5.3 Sentiment Analysis Results (CAMEL BERT)

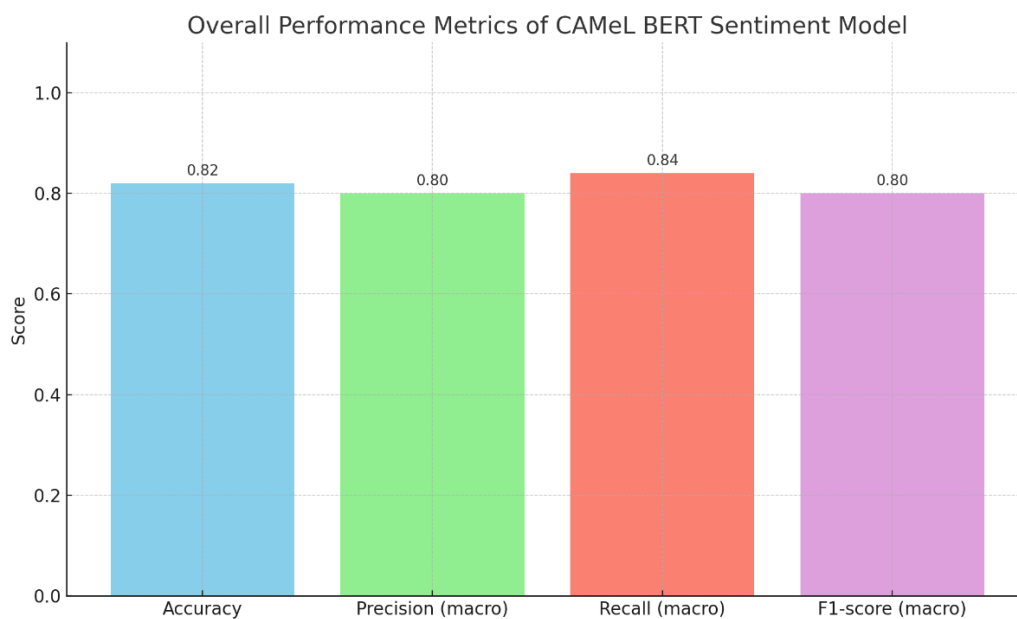
This is an excel file for the output of our nlp model

1	Title	Link	Date	Category	Content	Sentimen	Score	is_egx30
2	لبنك الأهلي	https://w	07 أبريل 25	أخبار الشركة	القاهرة مباشر	neutral	0.809787	غير مرتبط
3	عمومية يوتو	https://w	07 أبريل 25	أخبار الشركة	القاهرة مباشر	negative	0.624697	غير مرتبط
4	مدينة مصر ل	https://w	07 أبريل 25	التعاملات الد	القاهرة مباشر	positive	0.660371	غير مرتبط
5	عمومية الدول	https://w	07 أبريل 25	أخبار الشركة	القاهرة مباشر	positive	0.546495	غير مرتبط
6	مال الشركة	https://w	07 أبريل 25	أخبار الشركة	القاهرة مباشر	positive	0.625047	مرتبط
7	اليوم.. نهاية	https://w	07 أبريل 25	أخبار الشركة	القاهرة مباشر	positive	0.813987	مرتبط
8	عمومية إيديت	https://w	07 أبريل 25	جمعيات عم	القاهرة مباشر	positive	0.661635	غير مرتبط
9	عمومية "مار	https://w	07 أبريل 25	أخبار الشركة	القاهرة مباشر	negative	0.750608	غير مرتبط
10	مساهم يخف	https://w	06 أبريل 25	التعاملات الد	القاهرة مباشر	positive	0.529494	غير مرتبط
11	جي بي كورب	https://w	06 أبريل 25	أخبار الشركة	القاهرة مباشر	positive	0.720304	غير مرتبط
12	البورصة تعلن	https://w	06 أبريل 25	أخبار الشركة	القاهرة مباشر	positive	0.607689	غير مرتبط
13	عبر صفقة نق	https://w	06 أبريل 25	أخبار الشركة	القاهرة مباشر	positive	0.614662	غير مرتبط
14	تعيين ماهر	https://w	06 أبريل 25	أخبار الشركة	القاهرة مباشر	positive	0.781214	غير مرتبط
15	عمومية "بي	https://w	06 أبريل 25	أخبار الشركة	القاهرة مباشر	neutral	0.810847	غير مرتبط
16	تعيين محمد	https://w	06 أبريل 25	تعديلات مج	القاهرة مباشر	positive	0.758413	مرتبط
17	الدماركية	https://w	06 أبريل 25	اندماجات وا	القاهرة مباشر	positive	0.713261	غير مرتبط
18	مرسعودي	https://w	06 أبريل 25	أخبار الشركة	القاهرة مباشر	positive	0.631068	غير مرتبط
19	مصر الجديد	https://w	06 أبريل 25	أخبار الشركة	القاهرة مباشر	positive	0.639013	غير مرتبط
20	أرباح مصر ال	https://w	06 أبريل 25	نتائج أعمال	القاهرة مباشر	negative	0.38359	غير مرتبط
21	عمومية مطا	https://w	06 أبريل 25	أخبار الشركة	القاهرة مباشر	positive	0.784298	غير مرتبط
22	سير للتعليم	https://w	06 أبريل 25	أخبار الشركة	القاهرة مباشر	positive	0.748028	غير مرتبط
23	ليبيا للاستثما	https://w	06 أبريل 25	أخبار الشركة	القاهرة مباشر	positive	0.547352	مرتبط
24	اليوم.. نهاية	https://w	06 أبريل 25	أخبار الشركة	القاهرة مباشر	positive	0.676688	غير مرتبط
25	اليوم.. إيقاف	https://w	06 أبريل 25	أخبار الشركة	القاهرة مباشر	negative	0.77421	غير مرتبط
26	اليوم.. نهاية	https://w	06 أبريل 25	توزيعات نقد	القاهرة مباشر	negative	0.56179	غير مرتبط
27	عمومية غاز	https://w	06 أبريل 25	توزيعات نقد	القاهرة مباشر	positive	0.639047	غير مرتبط
28	وقت "CIB"	https://w	03 أبريل 25	بنوك ومالية	القاهرة مباشر	neutral	0.610176	مرتبط
29	جمعيات عم	https://w	03 أبريل 25	جمعيات عم	القاهرة مباشر	negative	0.606818	مرتبط

Evaluation Metrics

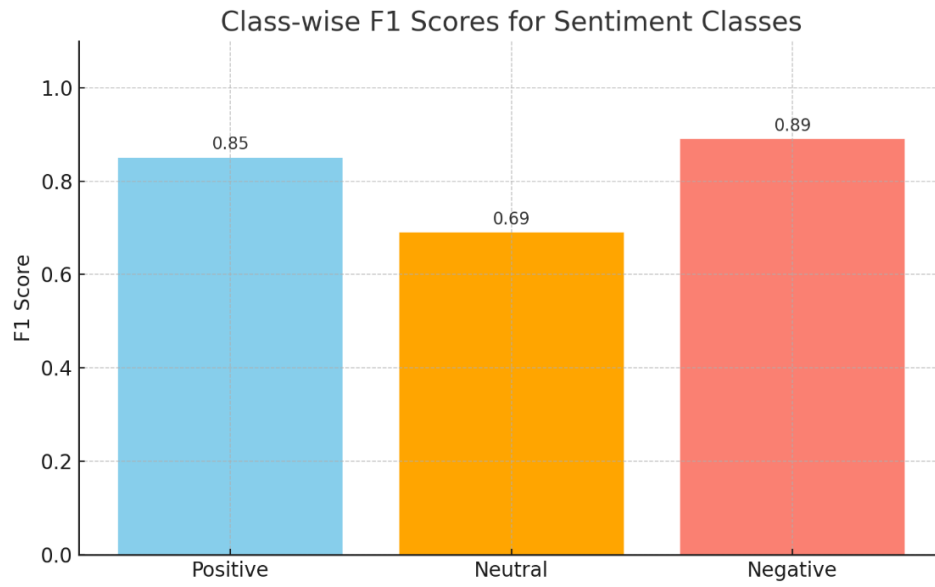
The fine-tuned CAMEL BERT model was evaluated on an 80/20 train-test split with stratification. Performance on the test set:

- **Accuracy:** %82
- **Precision (macro):** 0.80
- **Recall (macro):** 0.84
- **F1-score (macro):** 0.80

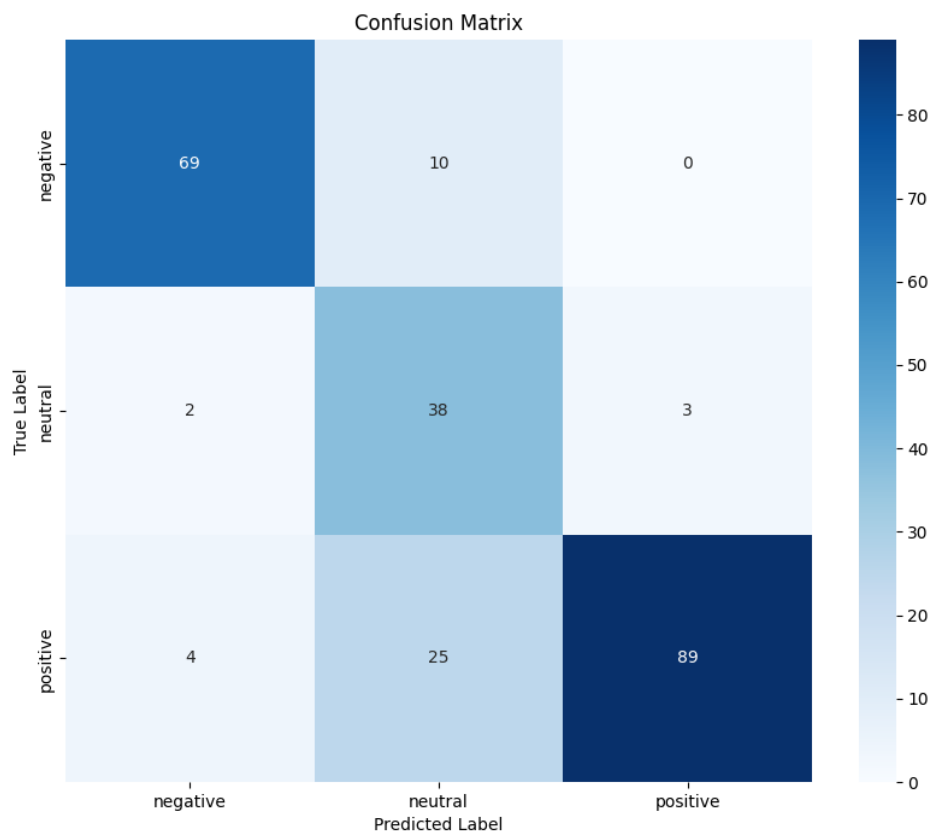


Class-wise Performance

- Positive: F1-score = 0.85
- Neutral: F1-score = 0.69
- Negative: F1-score = 0.89



Confusion Matrix



Note on Neutral Class Performance:

The **neutral** class shows the least accurate performance compared to the **negative** and **positive** classes. This is expected because the **CamelBERT** model was originally designed for binary classification (2 labels), and we manually modified its configuration to support 3 labels (negative, neutral, positive).

Additionally, the **neutral** class is particularly critical, as it can often lean toward either the **positive** or **negative** side depending on context. This makes it more challenging for the model to distinguish neutral instances accurately.

Insights and Observations

- Neutral news was most accurately predicted due to its volume.
- Positive and negative examples were slightly underrepresented but still well detected.
- Confidence scores correlated well with the number of financial keywords present in the text.

Preliminary Experiments and Challenges

- At first, a traditional **sentiment analysis model** was developed using basic Arabic lexicons, but the results were poor due to the complexity of financial language and the domain-specific meanings of common words.
- The team also experimented with **XGBoost** on top of extracted text features. However, this approach failed to model the relationship between unstructured news content and stock index behaviour

5.4 Forecasting Model Results (ARIMA, SARIMA, SARIMAX, LSTM, BiLSTM, Hybrid ARIMA–LSTM)

Evaluation Metrics

The following models were assessed based on their ability to forecast EGX-30 index prices. Standard regression metrics were used:

MODEL	RMSE	MAE	MAPE (%)
ARIMA (5,2,0)	23,977.15	20,227.87	68.69
ARIMA (3,2,3) – GRID SEARCH	2,909.38	2,541.47	8.54
SARIMA (5,2,0)(0,0,0)[0]	26,912.59	22,568.18	76.55
SARIMAX (WITH EXOGENOUS VARS)	8,790.66	7,976.33	27.17
LSTM REGRESSION	10,047.84	8,350.62	28.17
HYBRID ARIMA–LSTM	2,945.41	2,596.37	8.69
BIDIRECTIONAL LSTM	3,043.50	2,831.49	12.68

Performance Analysis

- **Hybrid ARIMA–LSTM** outperformed all other models across every metric. This model effectively corrected ARIMA's residual errors using an LSTM trained on technical indicators such as RSI, ATR, MACD, and Volume.
- **ARIMA (Grid Search)** also delivered solid performance when optimized, demonstrating the importance of parameter tuning.
- **SARIMAX**, which integrated exogenous features, significantly improved over plain SARIMA, validating the influence of technical indicators on price movement prediction.

- **Bidirectional LSTM** captured more complex dependencies than the unidirectional LSTM, leading to better forecasting accuracy.
 - **Standard ARIMA and SARIMA models** failed to capture the volatility and non-linearity of the Egyptian market, resulting in high errors.
-

Visual and Diagnostic Insights

- **Residual plots** revealed heteroscedasticity in ARIMA and SARIMA, which was better managed in LSTM-based models.
 - **Prediction curves** from BiLSTM and Hybrid models more closely tracked actual stock prices than their statistical counterparts.
 - **Loss convergence plots** confirmed that hybrid and BiLSTM architectures were more stable during training.
-

Experimental Challenges and Discarded Approaches

- A standalone **ANN model** for regression was tested but quickly abandoned. It lacked the temporal context needed for financial sequences and showed poor generalization.
- An attempt to use **LSTM alone for raw price regression** faced issues with market noise and unpredictability, producing erratic forecasts.



- While the **Hybrid ARIMA–LSTM** succeeded eventually, early versions suffered from synchronization issues between ARIMA residuals and LSTM input sequences.
- **Feature engineering** proved crucial. Omitting any major technical indicator (e.g., RSI or ATR) led to significant performance drops across all deep learning models.

5.5 Comparative Notes

Although the models were developed and evaluated independently, each offers a unique perspective that contributes to a comprehensive understanding of the EGX-30 market dynamics:

- **ARIMA and SARIMA** captured linear and seasonal trends but struggled in highly volatile conditions typical of emerging markets.
- **SARIMAX** improved significantly by incorporating technical indicators as exogenous variables, emphasizing the importance of feature enrichment in classical models.
- **LSTM** detected sequential patterns in technical indicators and responded well to strong directional movements.
- **Bidirectional LSTM** provided richer temporal insight by analyzing both past and future context, yielding improved performance over standard LSTM.

- **Hybrid ARIMA–LSTM** successfully combined the strengths of statistical forecasting and deep learning, outperforming individual models in most metrics.
- **CAMeL BERT**, operating on unstructured Arabic financial news, offered a sentiment-based interpretation of market sentiment and public reaction.
- **LSTM** captures numerical trends and volatility.

In future work, combining the outputs of these models—such as aligning sentiment scores with signal predictions—could produce a **multi-modal decision-support system** that fuses technical, statistical, and emotional data into unified investment insights.

5.6 Discussion

5.6 Discussion

This study explored six AI-driven forecasting and analysis models targeting the Egyptian stock market, specifically the EGX-30 index. The models addressed different dimensions of financial analysis—ranging from historical data forecasting to natural language sentiment interpretation:

Quantitative Technical Forecasting:

- **ARIMA, SARIMA, SARIMAX** modeled linear and seasonal structures in price behavior.
- **LSTM and Bidirectional LSTM** leveraged deep learning to extract nonlinear temporal dependencies from technical indicators.
- **Hybrid ARIMA–LSTM** merged statistical forecasting with deep residual learning to enhance robustness and accuracy.

Qualitative Sentiment Interpretation:

- **CAMeL BERT** was fine-tuned to classify Arabic financial news headlines into sentiment categories (Positive, Neutral, Negative), reflecting public perception and emotional tone.

Each model was trained, validated, and tested independently, producing insights from separate data modalities—numerical price trends and unstructured news content.

Key Observations:

- The **LSTM classifier** showed consistent signal accuracy in periods of clear market direction (e.g., sustained uptrends or corrections).
- The **CAMeL BERT model** maintained high classification performance even under imbalanced class distributions, due to careful data curation and domain-specific training.
- **Classical models** like SARIMA struggled with real-time volatility but became more effective with the integration of technical indicators (SARIMAX).
- **Bidirectional LSTM and Hybrid models** achieved the best forecasting accuracy, combining both contextual learning and residual correction.

By approaching the EGX-30 market from both statistical and deep learning angles—along with sentiment modeling—the study demonstrates that **multi-model strategies** can yield **richer and more actionable insights**, particularly in under-researched markets such as Egypt.

5.6 Summary

Several modelling strategies were tested and refined over time. Experiments using CNNs, LSTM-based regression, and hybrid LSTM-ARIMA approaches were found ineffective for EGX-30 forecasting and were discarded. In the NLP domain, lexicon-based methods and XGBoost classifiers also failed to capture financial semantics effectively.

Ultimately, the LSTM model achieved strong classification performance using engineered features from EGX-30 historical data. Meanwhile, the CAMeL BERT sentiment model showed high accuracy in classifying Arabic financial news. Both models demonstrate the feasibility of their respective approaches in the context of the Egyptian financial market

Chapter 6 conclusions and future work

6.1 Conclusion

This research successfully developed and evaluated two separate AI systems that analyze the Egyptian stock market from different dimensions:

The LSTM model utilized engineered technical indicators from EGX-30 data to provide actionable signal predictions.

The CAMeL BERT model was fine-tuned on Arabic financial news to deliver sentiment insights with high accuracy.

Both models confirmed the feasibility of AI-driven market analysis tailored to the Egyptian context and proved capable of handling local market data and Arabic content effectively.

6.2 Future Work

The current study evaluated a diverse set of forecasting models—ranging from traditional statistical methods (ARIMA, SARIMA, SARIMAX) to deep learning architectures (LSTM, Bidirectional LSTM, Hybrid ARIMA–LSTM). While each model was assessed independently, several future directions could enhance both their individual and collective utility:

- **Model Fusion and Integration:**

Develop a unified decision-support framework that fuses predictions from multiple models. For instance, aligning outputs from **Hybrid ARIMA–LSTM** with **CAMeL BERT sentiment scores** may offer stronger buy/sell signal confirmation than either model alone.

- **Advanced Ensemble Learning:**

Implement ensemble strategies (e.g., weighted voting or stacking) that combine ARIMA-based forecasts with neural models like LSTM and BiLSTM. This could increase robustness across different market regimes (e.g., trending vs. volatile).

- **Real-Time Multi-Model Deployment:**

Deploy selected models (e.g., SARIMAX for macro trends, BiLSTM for fine-grained signals) in a real-time streaming architecture for continuous updates on EGX-30 price movements and investor sentiment.

- **Model Generalization to Other Indices:**

Extend forecasting to **EGX70**, **EGX100**, or even regional indices (e.g., Saudi Tadawul, Dubai Financial Market) using the same modeling strategies. This would test the scalability and transferability of LSTM and ARIMA-based pipelines.

- **Multilingual & Multi-Source Sentiment Fusion:**

Improve the sentiment model by integrating **English-language** financial news and **social media feeds** alongside Arabic sources. Future versions of CAMEL BERT could be fine-tuned on cross-lingual corpora for broader semantic understanding.

- **Exploration of Transformer-Based Time Series Models:**

Investigate newer time-series architectures such as **Temporal Fusion**

Transformers (TFT) or **Informer**, which may outperform LSTM models in long-range dependency modeling.

By combining statistical rigor with deep learning flexibility, this project has built a strong foundation for intelligent financial forecasting in Arabic-speaking markets. Expanding and integrating these diverse models opens up a pathway for **multi-modal, real-time, and context-aware decision-support systems** tailored to complex and emerging financial ecosystems like Egypt's.

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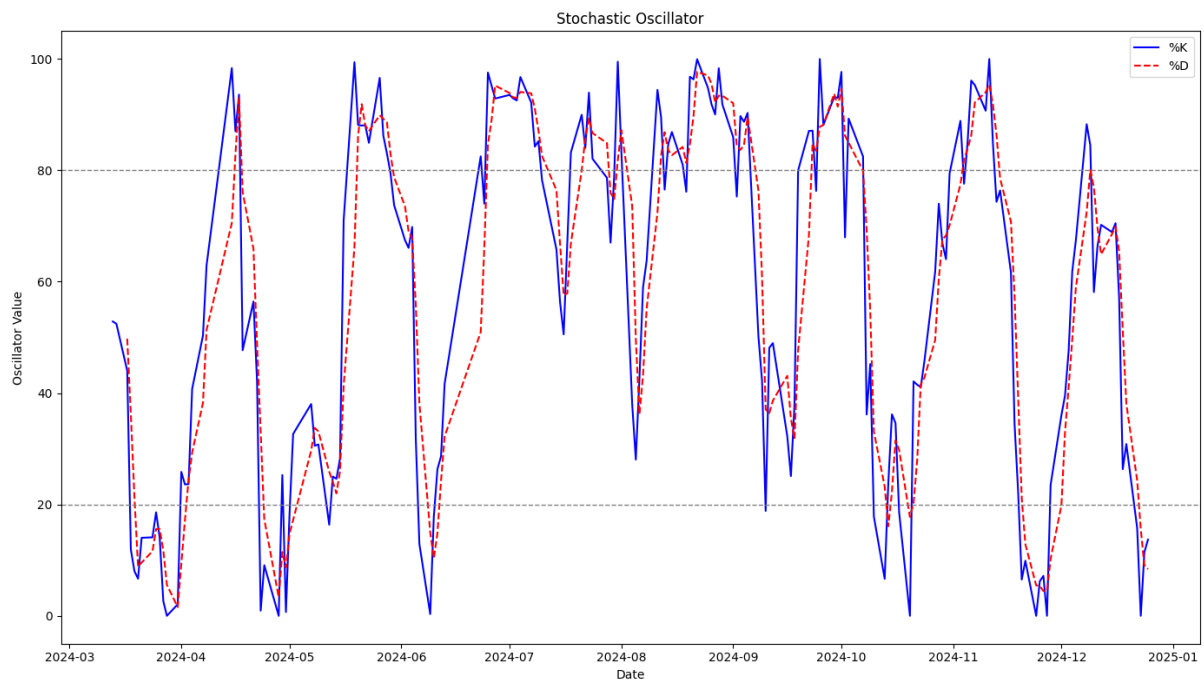
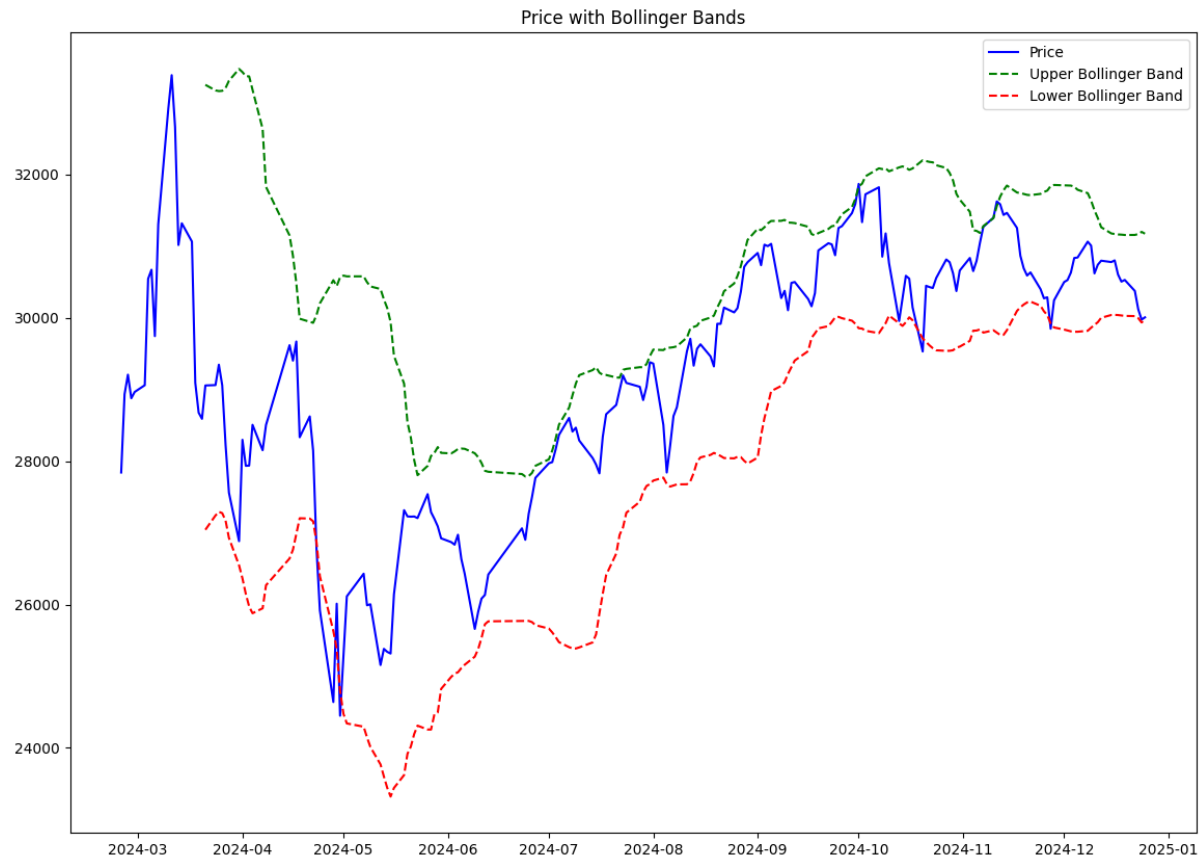
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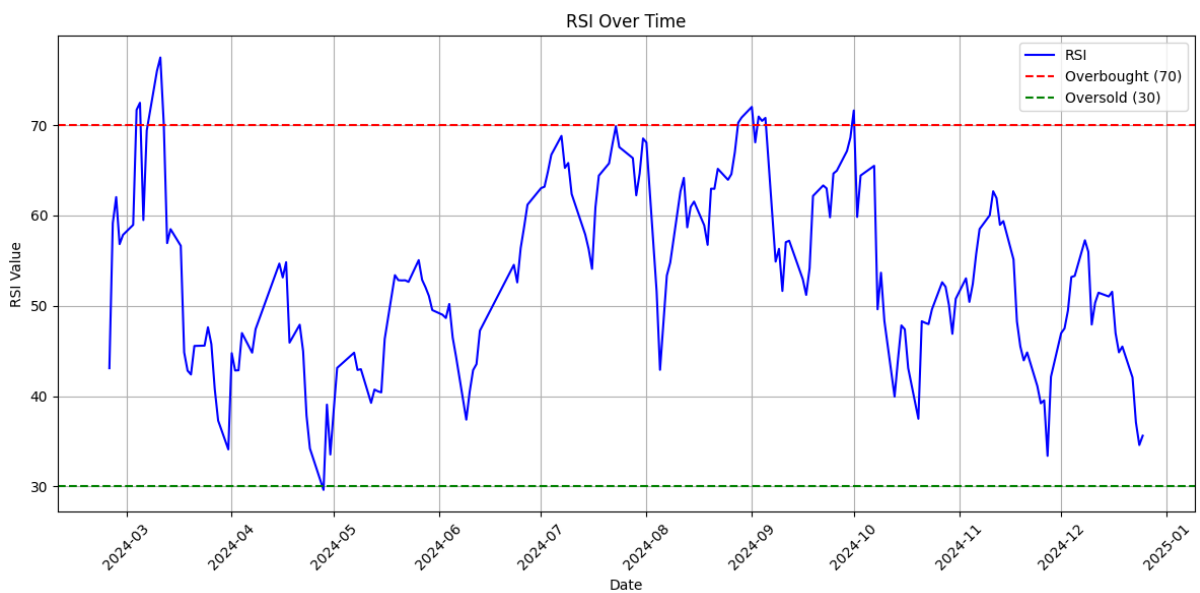
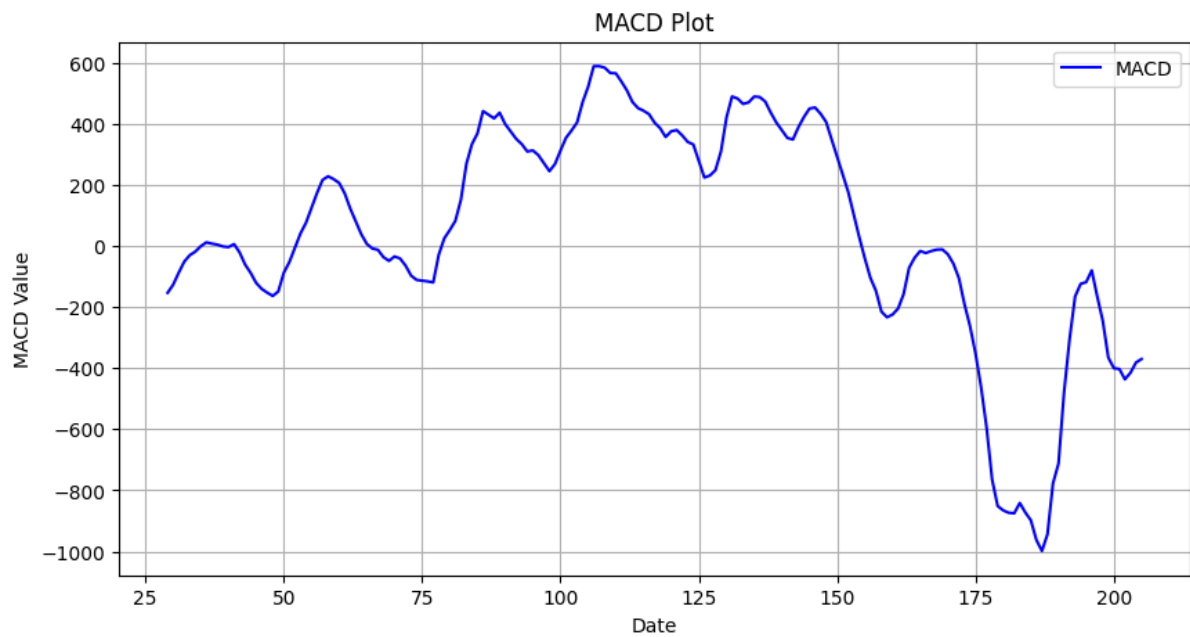
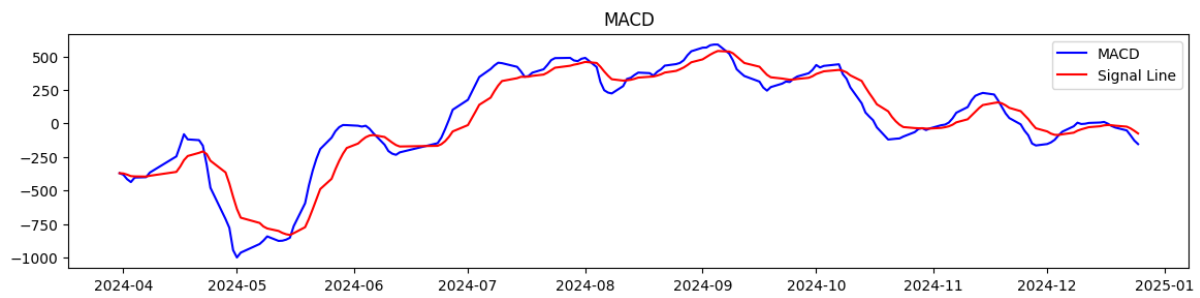
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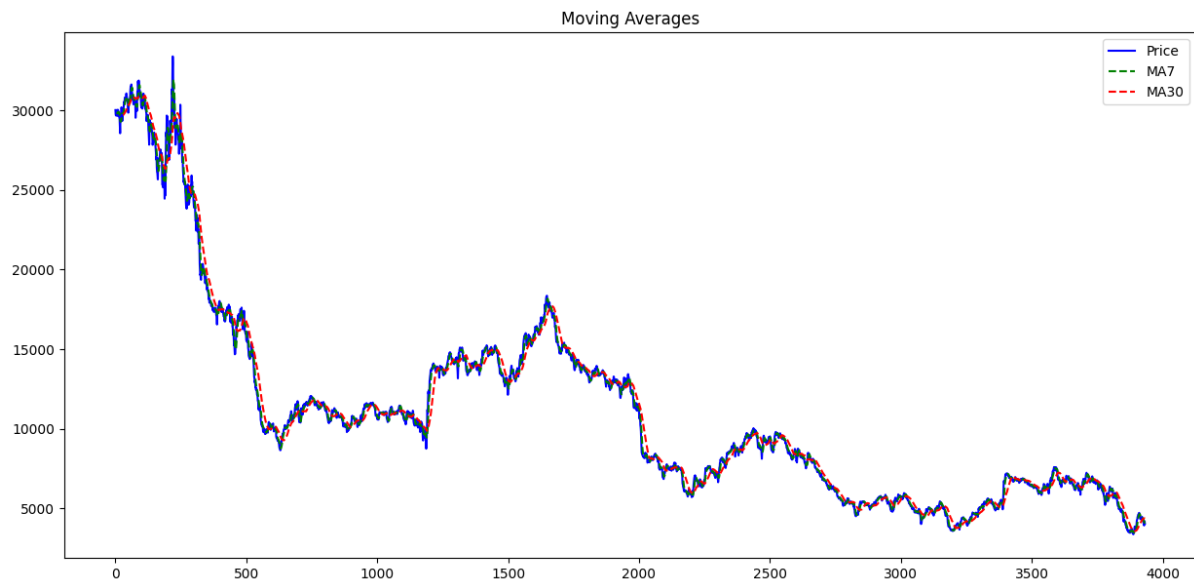
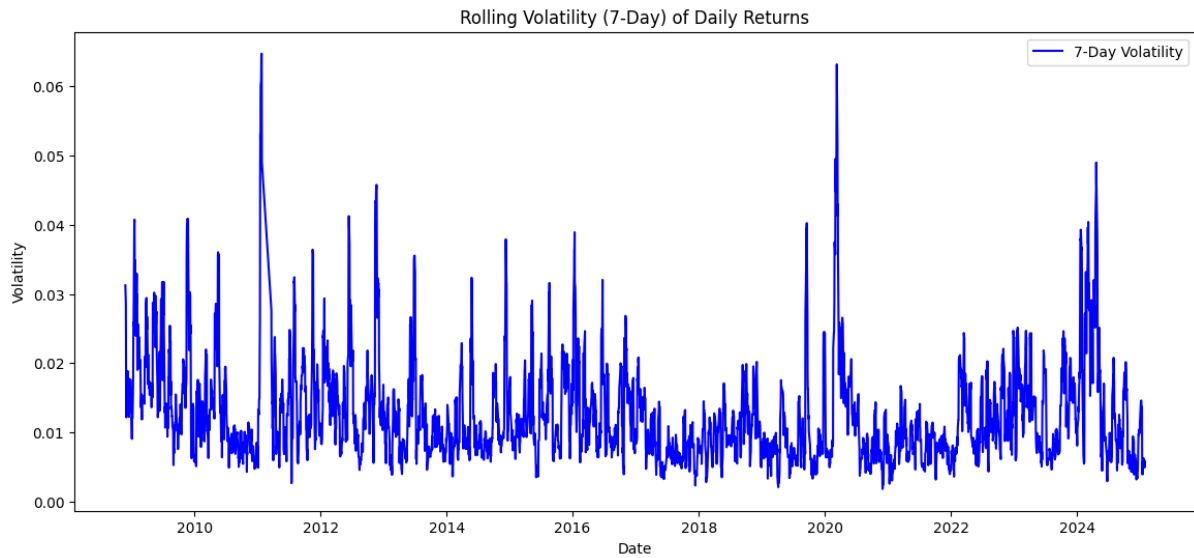
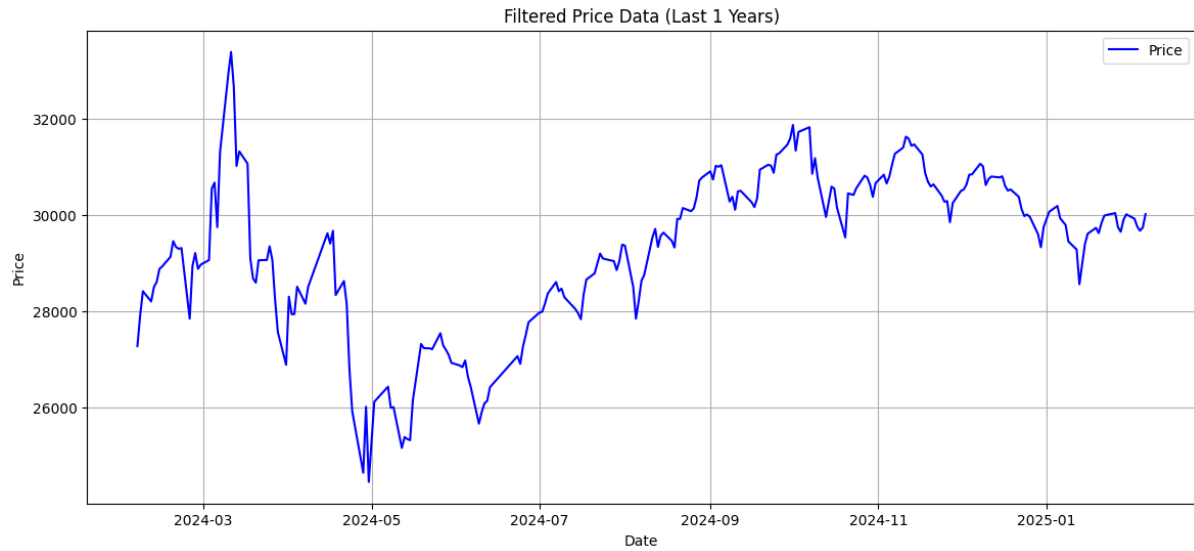
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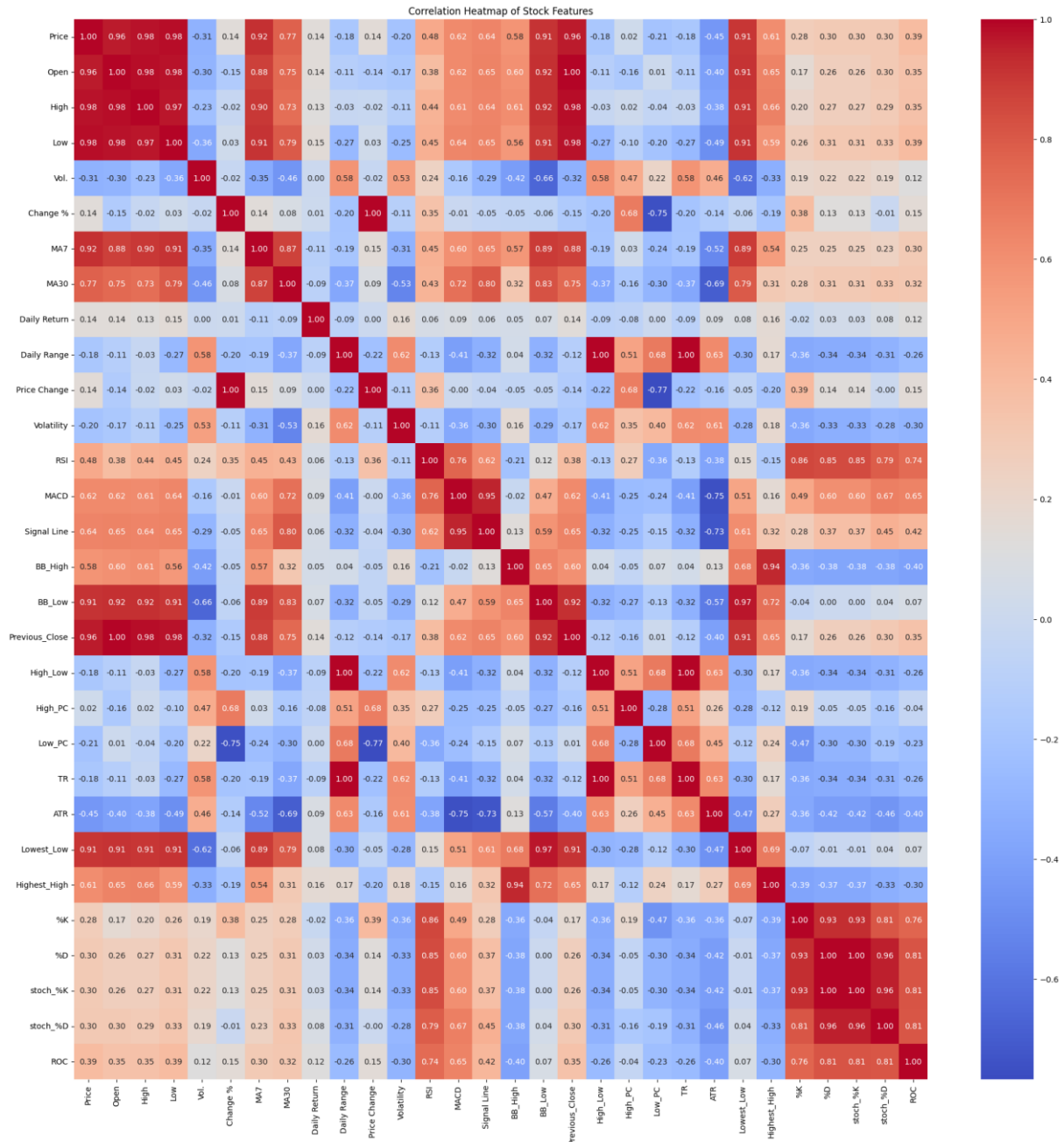


Appendices



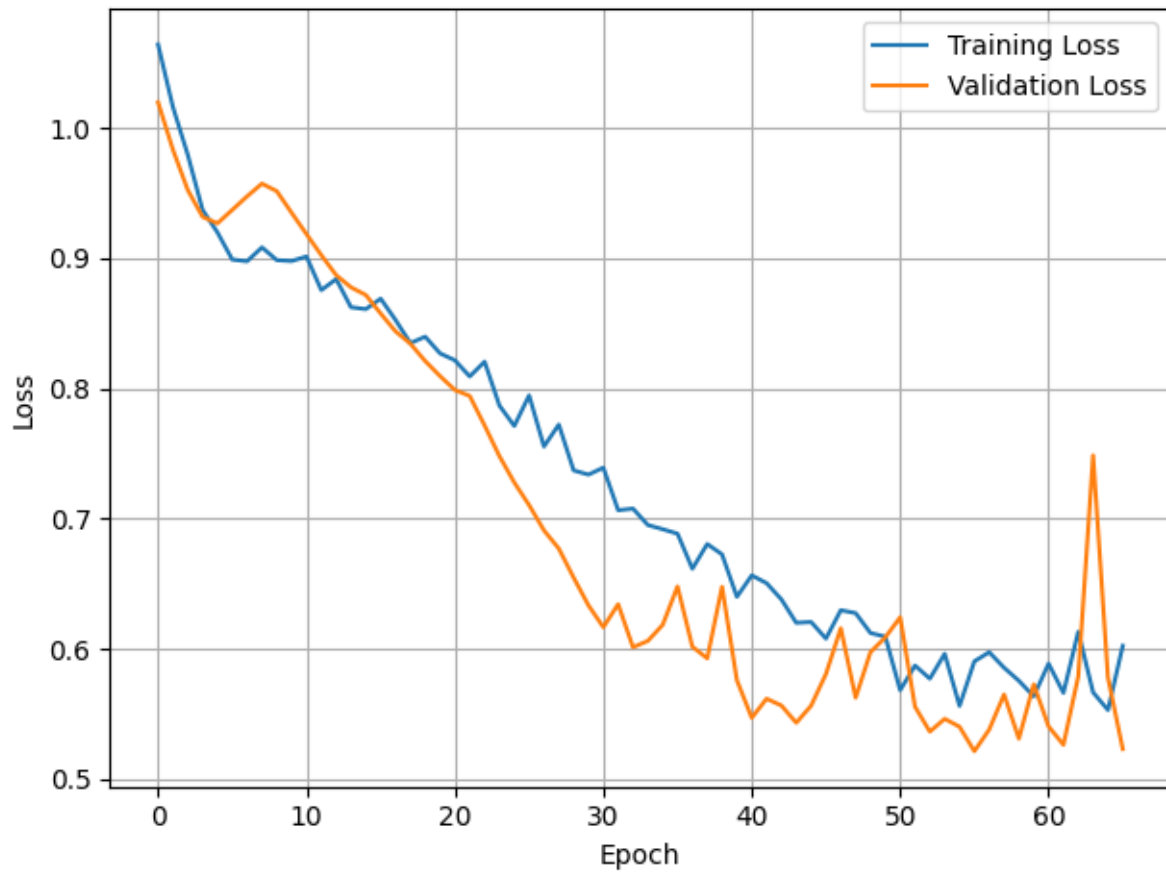




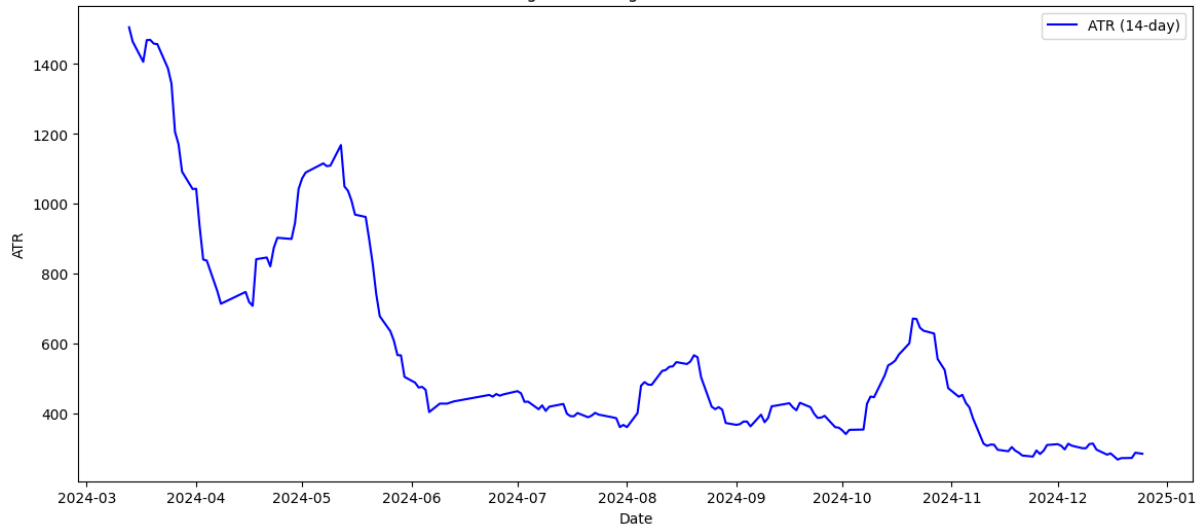




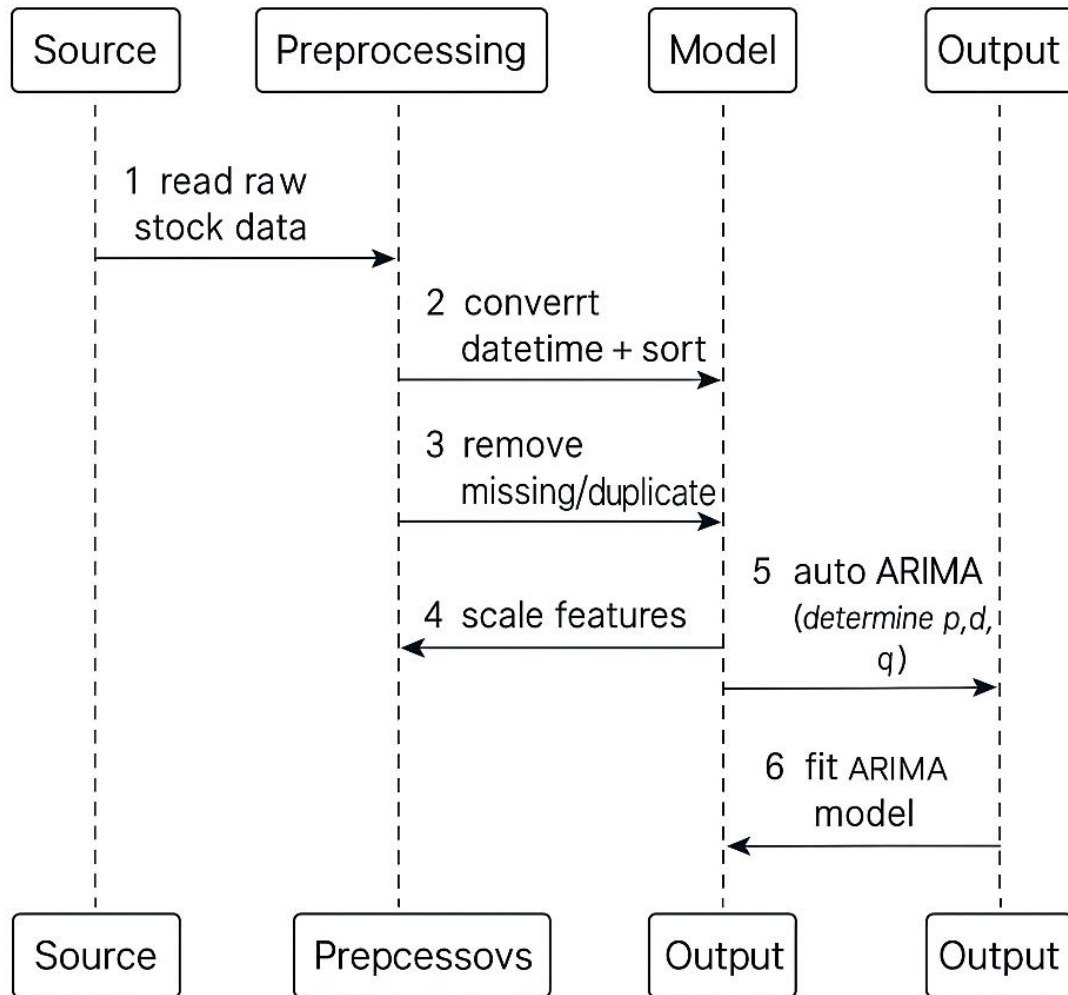
Loss Curve



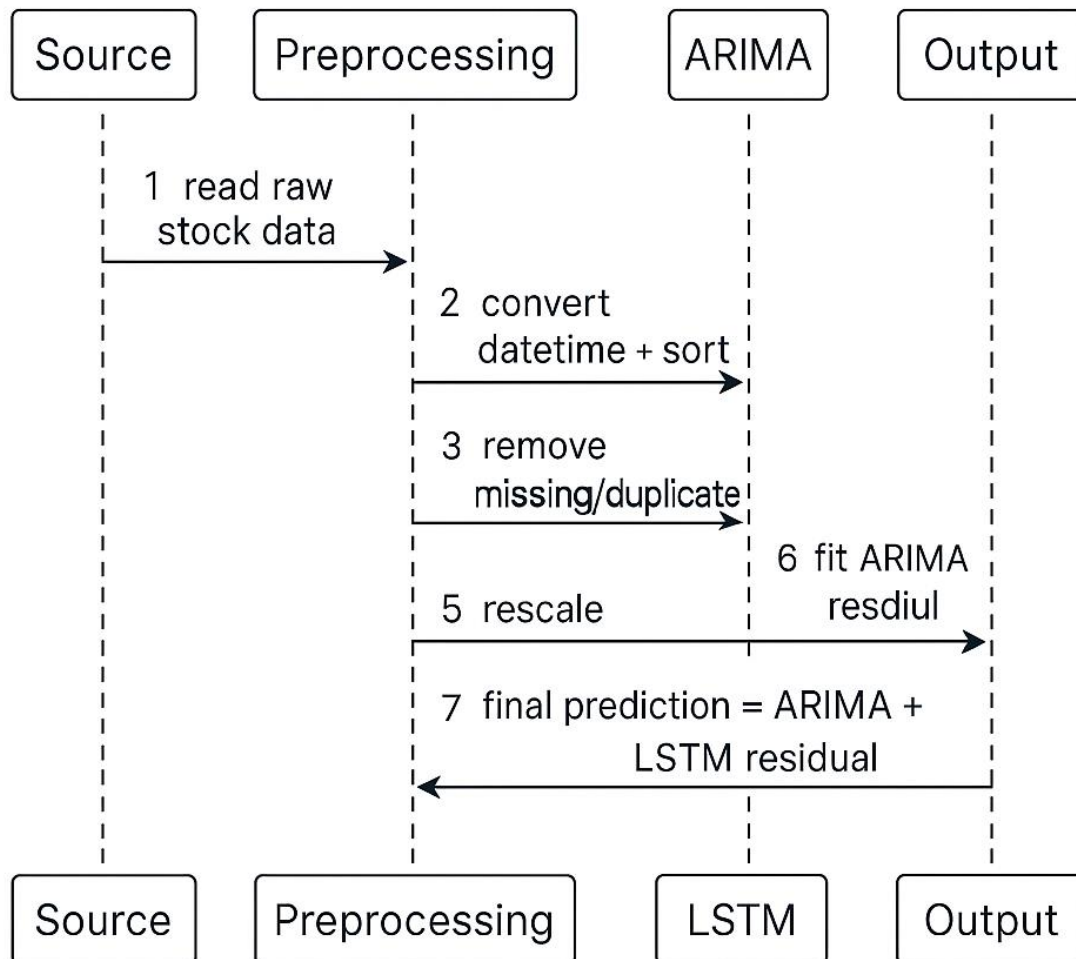
Average True Range (ATR) Over Time

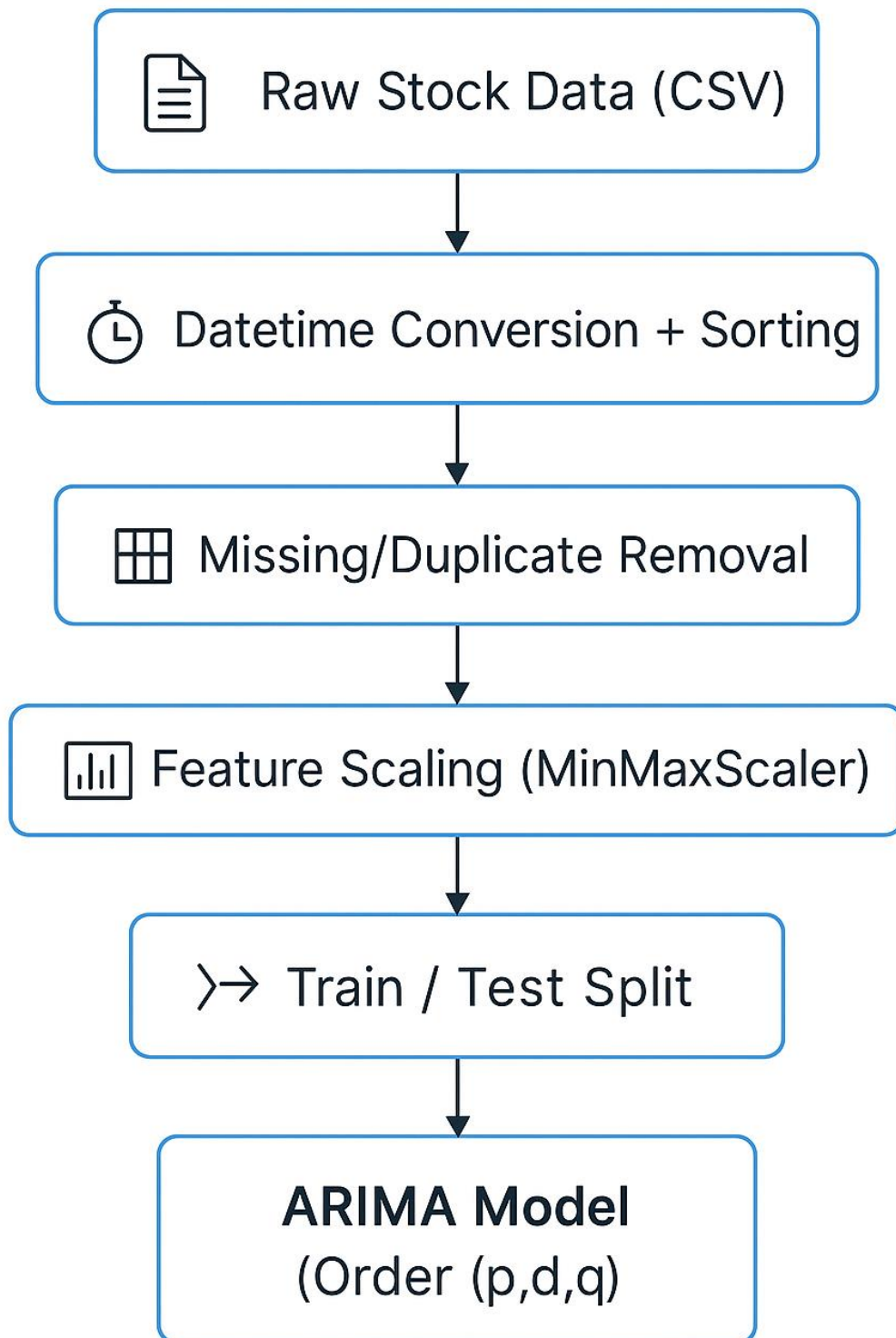


ARIMA



HYBRID MODEL





```
# تقسيم البيانات
train_df, test_df = train_test_split(df, test_size=0.2, random_state=42, stratify=df['label'])

# تحويل البيانات إلى تنسيق مناسب للتدريب
train_ds = Dataset.from_pandas(train_df)
test_ds = Dataset.from_pandas(test_df)

# تحويل النصوص إلى تنسيق مناسب للمودل
logging.info("معالجة البيانات...")
train_ds = train_ds.map(
    lambda x: preprocess_function(x, tokenizer),
    batched=True,
    batch_size=32,
    remove_columns=train_ds.column_names
)
test_ds = test_ds.map(
    lambda x: preprocess_function(x, tokenizer),
    batched=True,
    batch_size=32,
    remove_columns=test_ds.column_names
)

# تحويل البيانات إلى PyTorch tensors
train_ds.set_format(type='torch', columns=['input_ids', 'attention_mask', 'token_type_ids', 'labels'])
test_ds.set_format(type='torch', columns=['input_ids', 'attention_mask', 'token_type_ids', 'labels'])

# إعدادات التدريب
training_args = TrainingArguments(
    output_dir="./fin-sentiment-model",
    eval_strategy="steps",
    eval_steps=100,
    learning_rate=2e-5,
    per_device_train_batch_size=16,
    per_device_eval_batch_size=16,
    num_train_epochs=3,
    weight_decay=0.01,
    save_strategy="steps",
    save_steps=100,
    load_best_model_at_end=True,
    logging_dir="./logs",
    logging_steps=10,
    metric_for_best_model="f1",
    greater_is_better=True,
    report_to="none",
    fp16=True,
    gradient_accumulation_steps=2,
    warmup_steps=100,
    optim="adamw_torch"
)
```

```
training_args = TrainingArguments(  
    output_dir="./fin-sentiment-model",  
    eval_strategy="steps",  
    eval_steps=100,  
    learning_rate=2e-5,  
    per_device_train_batch_size=16,  
    per_device_eval_batch_size=16,  
    num_train_epochs=3,  
    weight_decay=0.01,  
    save_strategy="steps",  
    save_steps=100,  
    load_best_model_at_end=True,  
    logging_dir="./logs",  
    logging_steps=10,  
    metric_for_best_model="f1",  
    greater_is_better=True,  
    report_to="none",  
    fp16=True,  
    gradient_accumulation_steps=2,  
    warmup_steps=100,  
    optim="adamw_torch"  
)  
  
trainer = CustomTrainer(  
    model=model,  
    args=training_args,  
    train_dataset=train_ds,  
    eval_dataset=test_ds,  
    compute_metrics=compute_metrics  
)
```

```
# إعدادات التدريب
MODEL_NAME = "CAMEL-Lab/bert-base-arabic-camelbert-mix"
BASE_MODEL_PATH = "camelbert_base_model" # مسار حفظ النموذج الأساسي
TRAINED_MODEL_PATH = "trained_financial_sentiment_model"
DATASET_PATH = "TrainData.xlsx"

# تعريف mapping للـ labels
LABEL_MAPPING = {
    'negative': 0,
    'neutral': 1,
    'positive': 2
}
```

```
# التنبؤ
with torch.no_grad():
    outputs = model(**inputs)
    logits = outputs.logits
    # تحسين الثقة temperature scaling
    temperature = 0.3 # تقليل درجة الحرارة أكثر لزيادة الثقة
    scaled_logits = logits / temperature
    predictions = torch.softmax(scaled_logits, dim=-1)
    predicted_class = torch.argmax(predictions, dim=-1).item()
    confidence = predictions[0][predicted_class].item()

    # طباعة احتمالات كل تصنيف
    probs = predictions[0].tolist()
    logging.debug(f"احتمالات التصنيف: negative={probs[0]:.4f}, neutral={probs[1]:.4f}, positive={probs[2]:.4f}")

# تحليل الكلمات المالية
text_lower = text.lower()

# البحث عن الكلمات المالية في النص
positive_matches = []
negative_matches = []
```

```
# Build LSTM model
model = Sequential([
    LSTM(64, input_shape=(sequence_length, n_features), return_sequences=True),
    Dropout(0.3),
    LSTM(32),
    Dropout(0.3),
    Dense(32, activation='relu'),
    Dense(n_classes, activation='softmax')
])
```

```
# Train with validation data
history = model.fit(
    X_train, y_train,
    epochs=128,
    batch_size=32,
    validation_data=(X_val, y_val),
    callbacks=[early_stop],
    verbose=1
)
```

```
# Compile the model
model.compile(
    optimizer=Adam(learning_rate=0.0005),
    loss='categorical_crossentropy',
    metrics=['accuracy']
)
```

```
# Model summary
model.summary()
early_stop = EarlyStopping(
    monitor='val_loss',
    patience=10,
    restore_best_weights=True
)
```